

Mercer General Hospital – Emergency Department

CariSurg MedTech Pathways | Week 7 | Cost-Benefit Memo

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Terms used in this memo

- **Emergency Severity Index (ESI):** This is the 1-5 acuity scale that is assigned to a patient during triage, where ESI-1 is the most critical (life-threatening) and ESI-5 is the least urgent.
- **Recall (for ESI-1):** For all patients who were truly ESI-1, this lets the user know if the model correctly flagged the patient as critical. A recall of 0.582 means the model caught 58.2% of critical patients and missed the rest.
- **Macro F1:** This is the average performance score across all five ESI categories and treats each level equally important regardless of how many patients are assigned to it. Since ESI-1 patients are rare, macro f1 scores are used instead as accuracy alone rewards a model for getting the common predictions and not the less urgent cases.
- **Explainability:** How directly a clinician or an auditor can see why the model produced a particular result. “High” means the reasoning can be read off the model directly; “Medium” means it requires a secondary explanation and “Low” means the reasoning is obscure.

1. Verdict

Recommendation: Use the tuned XGBoost model as the primary model for the next development phase, while keeping logistic regression as an interpretable backup. Neither models should assign final ESI values as both should provide indicators to only support clinical decisions.

2. Recap

This memo evaluates and compares multiple machine learning models developed using the Yale EMLLC Emergency Department triage dataset, consisting of 55,121 patient encounters. After removing demographic variables and data-leakage features, 208 features remained. The dataset was then divided into training and testing sets using an 80/20 split, while maintaining the same distribution of ESI acuity levels in both groups to ensure a fair evaluation of model performance. During Week 5-6, two baseline models were used; a Logistic Regression and a Decision Tree. These models were assessed using accuracy and Macro F1 scores, which calculates the average F1 score across all 5 ESI Categories while ensuring less frequent classes received equal weighting.

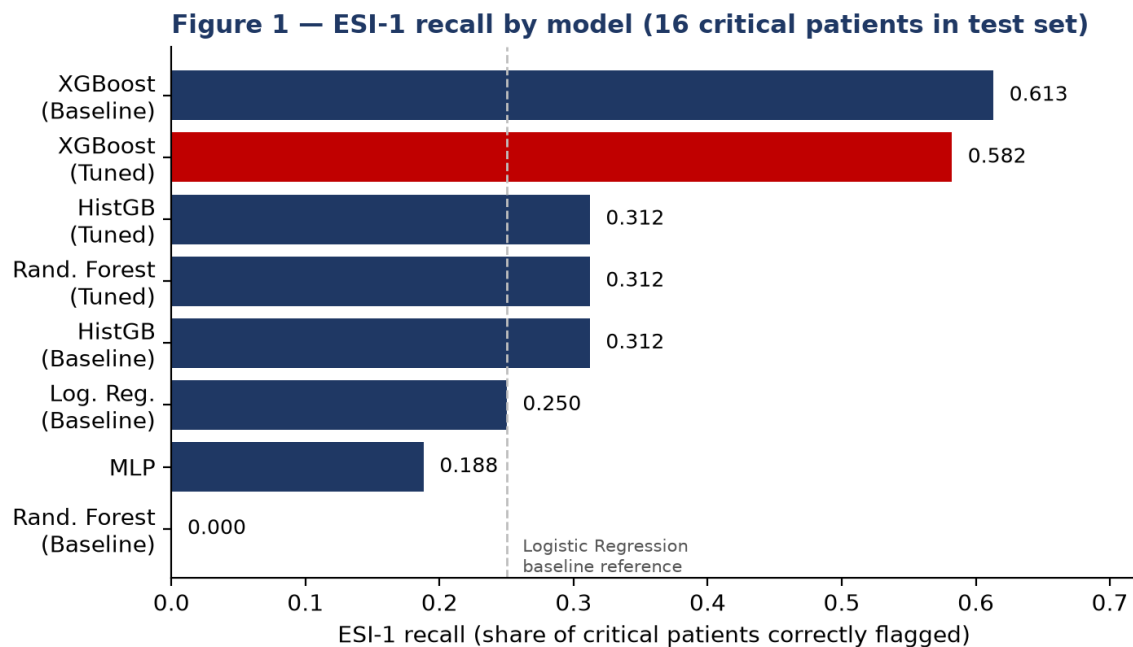
This week’s work expands the modelling approach through the introduction of physiological stress features which include shock index, pulse pressure and abnormal vital sign indicators. Two additional models were also implemented: XGBoost and a small multilayer perceptron (MLP) neural network. All models were evaluated using a six-axis benchmark framework consisting of accuracy, ESI-1 recall, Macro F1 score, training time, prediction time and explainability.

ESI-1 recall is the proportion of mostly critically ill patients correctly identified by the model. This was included as a separate evaluation metric because overall performance measures can overlook poor detection of high-risk cases. This additional feature ensures that reliable identification of these patients is the key clinical consideration in guiding the recommendation that is presented in this memo.

3. Benchmark Table (Six-Axis Comparison)

Model	Accuracy	ESI-1 Recall	Macro F1	Train Time (s)	Prediction Time (ms/patient)	Explainability
Logistic Regression (Baseline)	0.667	0.25	0.492	1.31	0.0017	High
XGBoost (Tuned)	0.668	0.582	0.463	9.64	0.0062	Medium
XGBoost (Baseline)	0.669	0.613	0.452	13.47	0.0081	Medium
MLP (Neural Network)	0.626	0.188	0.457	85.33	0.0055	Low
HistGradientBoosting (Tuned)	0.582	0.312	0.436	6.29	0.0086	Medium
HistGradientBoosting (Baseline)	0.536	0.312	0.404	6	0.0041	Medium
Random Forest (Tuned)	0.534	0.312	0.425	7.78	0.0241	Medium
Random Forest (Baseline)	0.641	0	0.392	13.34	0.0985	Medium

The test set contains only 16 ESI patients out of 11,025 total cases. As a result, ESI-1 recall is based on a very small number of patients, meaning that the classification of a single patient can substantially affect the reported recall



4. Three Arguments for XGBoost (Tuned)

- a. Much stronger detection of most critical patients: An ESI-1 recall of 0.582 means the model correctly identifies nearly 58% percent of patients in comparison with the logistic regression baseline of 0.25. This is the most clinically important metric for ED triage.
- b. Balanced improvement in safety and overall performance: Despite the increase in the ESI-1 recall, the model maintains an accuracy of 0.668 and a Macro F1 score of 0.463, both close to the Logistic Regression Baseline. This indicates an improvement in detecting critical patients is achieved without any drop in overall performance.
- c. Efficient enough for real-time clinical use: The model is trained in 9.64 seconds and requires 0.0062 milliseconds per patient for prediction; this makes it well suited for ED workflows. It also more efficient compared to the XGBoost baseline while delivering better clinical performance.

5. Three Arguments Against XGBoost

- a. Reduced explainability: Compared with the Logistic Regression, the explainability decreases from High to Medium. With Logistic Regression, clinicians can easily interpret each vital signs influences a prediction. XGBoost is a more complex model, therefore, additional software is needed to explain why the model assigned a patient to a particular ESI level.
- b. Critical cases are still missed: Although the model improved the ESI-1 recall to 0.582, the model still fails to identify most critical cases. The model should therefore be used as a decision support tool that provides clinicals with a second opinion.
- c. Improving overall performance required a small trade-off: The XGBoost baseline achieved a higher ESI-1 recall than the tuned version. This is because the tuning was designed to improve the model's performance across all five ESI levels rather than only focusing on identifying the most critical patients.

6. Risks and Recommendation

- ESI-1 recall is based on only 16 patients in the test set, meaning the results have a high level of uncertainty. A larger test sample could change the performance between the XGBoost (Tuned) and XGBoost (Baseline).
- No model was specifically optimised to maximise ESI-1 detection. Future work could explore any approaches that can prioritise critical patient identification while maintaining overall performance.

Recommendation

Proceed with the XGBoost (Tuned) as the primary model for Week 8 development, while keeping logistic regression as a comparison mode. Complete the ongoing ESI-1 error analysis before any pilot deployment. Both models should continue to provide risk indicators only, supporting clinical judgement rather than recommending ESI levels directly.